Pembroke Hill Invitational

12/6/2025

Circuit Lab Test

Division B

1. A circuit consists of a 12V battery connected to a network of resistors. Resistor A (4 ohms) is in series with a parallel combination of Resistor B (6 ohms) and Resistor C (3 ohms). What is the total current flowing out of the battery? [2 pts]

2. In the circuit described in Question 1, what is the voltage drop across Resistor B? [2 pts]

3. How much power is consumed by Resistor C in the circuit from Question 1? [2 pts]

4. A battery has an electromotive force (EMF) of 9.0V. When connected to an external load of 10 ohms, the terminal voltage drops to 8.5V. What is the internal resistance of the battery? [2 pts]

5. Two capacitors, C1 = 100 microF and C2 = 300 microF, are connected in parallel to a 12V source. What is the total charge stored in the system? [2 pts]

6. Two resistors, R1 = 1k ohms and R2 = 4k ohms, are in series across a 15V source. What is the voltage across R2? [2 pts]

7. A 100W light bulb is left on for 24 hours. If electricity costs $0.15 per kilowatt-hour (kWh), how much did it cost to run this bulb? [2 pts]

8. True or False: [1 pt]
   Conventional current flows in the same direction as the actual electron flow in a copper wire.

9. If the distance between two point charges is tripled, how does the electrostatic force between them change according to Coulomb's Law? [1 pt]

   A) Increase by a factor of 9
   B) Decrease by a factor of 9
   C) Increase by a fator of 3
   D) Remain constant

10. You are looking directly at the end of a solenoid (coil of wire). If the current flows clockwise around the loops from your perspective, which magnetic pole is facing you? [1 pt]

11. A transformer has 500 turns on the primary coil and 50 turns on the secondary coil. If the input voltage is 120V AC, what is the output voltage? [2 pts]

12. An ideal Ammeter should have what internal resistance? [1 pt]

   A) Zero
   B) Infinite
   C) Matching the circuit
   D) None of the above

13. An ideal Voltmeter should be connected in which configuration with the component it is measuring? [1 pt]

14. A straight wire carries current vertically upwards. If you view the wire from above (looking down), in which direction are the magnetic field lines oriented? [1 pt]

15. Who is historically famous for championing the Direct Current (DC) standard during the "War of the Currents"? [1 pt]

16. Which device is specifically designed to protect humans from electrical shock in wet areas by detecting imbalances between live and neutral wires? [1 pt]

17. In a P-N junction diode, current only flows significantly when the anode is at a _____ potential than the cathode. [1 pt]

18. In a standard alkaline battery or voltaic cell, where does oxidation occur: [1 pt]
    A) Anode
    B) Cathode

19. When you rub a glass rod with silk, the glass becomes positively charged. Did the glass - [1 pt]
    A) Gain protons
    B) Lose electrons

20. An electron moves horizontally east through a magnetic field pointing vertically down. In which direction is the magnetic force on the electron? [1 pt]
    A) North
    B) South
    C) East
    D) West

21. What electrical component is represented by a zigzag line with an arrow drawing through it diagonally? [1 pt]

22. A string of holiday lights is wired in series. One bulb burns out (filament breaks). What happens to the other bulbs? [1 pt]

23. Which digital logic gate outputs TRUE (1) only if both inputs are TRUE (1)? [1 pt]

24. You measure the resistance of a component and your multimeter reads 0.1 ohms. Is this component likely Open or Shorted? [1 pt]

25. To control a single hallway light from two different switches, what specific type of switches are required? [1 pt]

A) SPST
B) SPDT
C) DPDT

26. In a schematic, if two wires cross *with* a dot at the intersection, then they electrically _____. [1 pt]

27. You use a low-quality voltmeter with low internal resistance to measure the voltage across a high-value resistor. Will your measured reading be - [1 pt]

A) Higher
B) Lower
C) Same as the actual theoretical voltage

28. In an LED, what does the shorter leg and longer leg indicate? [2 pts]

29. A fuse is rated for 5A. If you run a 600W device on a 120V circuit protected by this fuse, will it blow? [2 pts]

30. True or False: [1 pt]
Breaking a bar magnet in half results in one isolated North magnet and one isolated South magnet (monopoles)

31. According to Kirchhoff's Current Law (KCL), the sum of currents entering a junction equals the sum of currents leaving the junction. This law is based on which fundamental physics principle? [1 pt]

A) Conservation of Energy
B) Conservation of Momentum
C) Conservation of Charge
D) Ohm's Law

32. If the voltage across a resistor is doubled, and the resistance is simultaneously halved, how does the power dissipated by the resistor change? [1 pt]

A) It increases by a factor of 2
B) It increases by a factor of 8
C) It increases by a factor of 4
D) It increases by a factor of 16

33. Which of the following correctly defines the unit of electric potential, the Volt? [1 pt]

A) Coulomb per Second
B) Watt per Ampere
C) Ampere per Second
D) Joule per Coulomb

34. Which statement is true regarding an ideal transformer? [1 pt]

A) It can convert DC voltage to AC voltage
B) Power in the primary equals power in the secondary
C) It changes voltage but not current
D) It works effectively with DC sources

35. Which of the following factors does not influence the strength of the magnetic field produced by a simple electromagnet? [1 pt]

A) The resistance of the wire material

B) The type of core material used

C) The current flowing through the wire

D) The number of coils of wire

Hands on Parts:

36. Build a circuit with 2 Resistors in Parallel connected in series to two LEDs in Parallel. [2.5 pts]

37. Build a circuit with 2 Resistors and 2 LEDs in series. [2.5 pts]

38. Measure the voltage across two points on the breadboard using the provided multimeter and record it with correct units. [5 pts]

39. Measure the current across two points on the breadboard using the provided multimeter and record it with correct units. [5 pts]

Pembroke Hill Invitational - Repaired practice edition

Division B | Circuit Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

35. (1 points) Authored replacement for source item 35: For a simple solenoid with fixed current and fixed core geometry, does increasing the number of turns per unit length increase or decrease its magnetic field? Explain.

Answer: _____
_____
_____

repair-case. (2 points) Authored written substitute for unavailable source material: Equal LED current guarantees equal brightness?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

A. No

B. Always

C. Only for any red LED

D. Only at 5 V

Answer: _____